

A. Ambispective Case Study: NCT02566993

Retrospective Analysis

Prospective Simulation

	Frequency	Frequency in Trials with Chemotherapy Drugs	Average Start Date (Year ± Month)	EPSM	Exclusionary Rate (Overall)	Exclusionary Rate (18-50, 50-65, >65)	Exclusionary Rate (Female, Male)	Exclusionary Rate (White, Asian, African American)
E-Second invasive malignancy (chemo/radio)	5.3%	5.8%	2009.0 ± 6.97	0.98	83.9%	84.7% 85.4% 83.4%	85.2% 82.1%	83.5% 69.4% 87.8%
E-Interstitial lung disease or pulmonary fibrosis	12.9%	13.8%	2014.4 ± 4.42	2.08	71.1%	55.9% 68.7% 72.8%	71.9% 70.2%	70.6% 56.9% 76.2%
E-Infection due to external drainages	4.9%	5.4%	2010.4 ± 6.98	1.74	33.3%	34.7% 33.0% 33.2%	41.9% 21.7%	34.3% 23.6% 30.4%
I-Hemoglobin ≥ 9.0 g/dL	3.9%	5.8%	2010.9 ± 6.06	1.64	25.7%	25.4% 24.8% 26.1%	15.8% 39.2%	28.1% 18.1% 17.3%
E-Pregnant patients	0.6%	0.7%	2008.9 ± 3.67	0.48	18.1%	35.6% 21.7% 16.0%	16.3% 0.0%	17.1% 19.4% 22.4%
E-Immunodeficiency including HIV infection	15.7%	18.5%	2014.4 ± 5.06	1.65	17.8%	25.4% 14.0% 18.3%	16.2% 19.9%	20.2% 6.9% 9.0%
I-Absolute neutrophil count ≥ 2.0×10 ⁹ /L	1.2%	1.8%	2007.8 ± 5.46	2.44	17.3%	17.8% 14.2% 18.2%	10.0% 27.2%	17.4% 11.1% 18.2%
I-Normal LVEF (ECHO/MUGA)	0.2%	0.0%	2013.2 ± 0.00	0.12	13.5%	22.0% 19.3% 21.8%	11.3% 34.6%	20.8% 22.2% 23.4%
I-ECOG PS ≤ 2	10.2%	8.0%	2011.2 ± 6.07	1.03	13.0%	8.5% 12.2% 13.5%	22.2% 0.8%	11.2% 11.1% 21.4%
I-Albumin ≥ 3.0 g/dL	0.2%	0.4%	1999.8 ± 0.00	0.26	12.1%	14.4% 15.7% 18.3%	12.2% 24.8%	19.8% 12.5% 8.8%
I-Creatinine clearance ≥ 30 mL/min	1.4%	0.4%	2015.7 ± 3.03	0.25	11.9%	20.3% 17.7% 20.6%	11.6% 31.2%	21.4% 9.7% 15.8%
I-Total bilirubin ≤ 1.5×ULN	6.7%	8.0%	2008.4 ± 6.21	1.95	10.4%	15.3% 11.1% 11.9%	7.6% 17.7%	13.8% 9.7% 3.9%
I-Adequate organ function within 7d	2.9%	4.0%	2007.1 ± 4.84	2.71	7.8%	10.2% 7.8% 7.7%	10.7% 4.0%	7.2% 6.9% 10.9%
I-AST ≤ 3×ULN	0.6%	0.4%	2014.6 ± 3.10	0.30	7.6%	8.5% 6.7% 8.7%	4.7% 13.1%	9.8% 6.9% 1.7%
I-ALT ≤ 3×ULN	0.8%	0.7%	2013.4 ± 3.31	0.29	7.1%	6.8% 5.5% 7.5%	4.2% 11.0%	8.4% 1.4% 2.2%
E-Oxygen requirement within 2 weeks	0.2%	0.4%	2016.7 ± 0.00	0.09	6.2%	5.1% 7.1% 6.1%	9.6% 1.7%	5.8% 1.4% 9.0%
E-Prior PM01183/topotecan/anthracycline	1.0%	1.4%	2011.4 ± 4.53	0.10	4.9%	5.9% 5.3% 4.8%	6.7% 2.6%	5.0% 1.4% 5.1%
I-Platelets ≥ 100×10 ⁹ /L	10.4%	14.9%	2006.5 ± 6.14	2.28	4.3%	14.4% 11.5% 11.6%	15.8% 6.3%	11.6% 16.7% 11.4%
E-Hepatitis B w/ positive (HBsAg+, PCR+)	1.0%	1.1%	2014.5 ± 3.70	0.36	4.2%	7.6% 4.4% 3.9%	2.5% 6.5%	4.6% 4.2% 2.4%
E-Hepatitis C w/ positive (Ab+, PCR+)	0.6%	0.7%	2017.7 ± 0.73	3.23	3.8%	2.5% 4.0% 3.8%	3.2% 4.6%	3.9% 5.6% 2.9%
E-Prior malignancy >3 yrs remission	3.7%	2.9%	2014.6 ± 7.17	0.30	3.5%	2.5% 4.0% 3.5%	4.0% 2.9%	3.7% 1.4% 3.2%
I-Creatine phosphokinase ≤ 5×ULN (disease)	0.2%	0.4%	2007.0 ± 0.00	0.16	2.5%	3.4% 1.3% 2.8%	1.4% 4.0%	2.4% 0.0% 3.4%
I-SCLC (post-platinum, ≥30d CFI)	2.2%	3.6%	2014.4 ± 7.18	1.94	2.3%	9.3% 9.5% 10.2%	5.5% 16.1%	11.0% 6.9% 6.3%
I-SCC-unknown primary, Ki-67>50%	0.2%	0.4%	2016.7 ± 0.00	0.09	2.2%	12.7% 8.6% 13.0%	5.9% 20.3%	12.1% 11.1% 12.2%
E-More then 1 prior chemo regimen	6.7%	6.5%	2011.2 ± 5.63	0.54	2.0%	3.4% 2.7% 1.7%	2.4% 1.4%	2.0% 1.4% 1.9%
I-Written informed consent	24.5%	24.6%	2010.9 ± 5.99	4.01	1.4%	4.2% 4.0% 4.8%	3.3% 6.3%	5.1% 4.2% 2.4%
E-Pending palliative RT/surgery (<4w)	2.5%	2.9%	2010.6 ± 5.70	2.61	1.1%	0.0% 1.8% 1.0%	0.7% 1.7%	1.3% 1.4% 0.0%
E-Compliance risk	11.9%	14.1%	2010.4 ± 5.70	0.96	0.6%	0.0% 0.7% 0.7%	0.5% 0.8%	0.6% 0.0% 0.7%
E-Uncontrolled arrhythmia	1.0%	1.8%	2003.1 ± 2.15	1.01	0.5%	0.0% 0.2% 0.6%	0.3% 0.8%	0.6% 0.0% 0.2%
I-Whole brain radiotherapy (weeks)≥4w	1.4%	1.8%	2005.9 ± 5.32	2.92	0.2%	0.0% 0.0% 0.2%	0.1% 0.3%	0.2% 0.0% 0.0%
I-Male contraception agreement	17.4%	18.8%	2010.2 ± 7.24	2.16	0.1%	0.0% 0.2% 0.1%	0.2% 0.0%	0.2% 0.0% 0.0%
I-Prophylactic cranial irradiation (weeks) ≥2	0.4%	0.7%	2013.7 ± 2.92	0.06	0.1%	0.0% 0.0% 0.1%	0.1% 0.1%	0.1% 0.0% 0.0%

B. Rules Optimization

Before Optimization

During Optimization

After Optimization

